

Neural Field Perceiver v14

Phase-1 Architecture Spec for Cross-Patient Speech Decoding

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This document describes the active **Phase-1** v14 design only. It folds in the **B-1 amendment** of 2026-04-16 late, which replaced the per-parcel pooled-token interface with a per-electrode token interface plus a soft parcel embedding, and replaced factored spatial-then-temporal attention with combined spatiotemporal attention. The amendment record at `docs/v14_core_contract_amendment.2026-04-16.md` is the authoritative source for the rewrite rationale; the per-blocker status log at `docs/implementation_tasks.md` remains the source of truth for implementation gating.

1 Problem and Scope

Each patient’s μ ECoG array is a sparse sample of cortical high-gamma activity during non-word repetition. Arrays are placed by the surgeon, not by protocol. Channel indices therefore have no shared meaning across patients.

Phase 1 asks one concrete question:

Can a fixed-atlas, shared model decode the 3-phoneme sequence of each trial across patients, using parcel identity as the only cross-patient anchor and without learned per-patient calibration?

The task is the PhonemeSequence non-word repetition dataset:

- 52 canonical tokens, each 3 phonemes long
- 9 phoneme classes (AA, AE, B, G, IY, K, P, UW, V)
- one sample = one response-onset-locked trial
- input window: $t_{\min} = -0.5\text{ s}$, $t_{\max} = 1.0\text{ s}$
- signal: z-scored high-gamma at 200 Hz

Phase-1 training and evaluation are **left hemisphere only**. Core patients are S14, S26, S33, and S62. Extended left-hemisphere patients S16, S23, and S39 audit in parallel. Right-hemisphere patients (S22, S58) and cross-modality joins are deferred to Phase 2.

2 Design Principle

v14 splits the problem into two parts:

1. **Spatial calibration** (per-patient, physics-constrained): patient-specific electrodes are placed in a shared atlas via a fixed Brainnetome surface parcellation on fsaverage. The atlas does $\approx 90\%$ of calibration; learned refinement is deferred to Phase 2.

2. **Shared dynamics** (cross-patient, unconstrained): a small relational-temporal transformer with an autoregressive decoder reads per-electrode tokens annotated with a soft parcel-identity embedding.

The shared interface is a per-electrode token stream

$$\mathbf{z} \in \mathbb{R}^{B \times (N_e \cdot T) \times d}, \quad (1)$$

with

- N_e = patient-specific number of non-artifact electrodes (128 or 256 for the Phase-1 cohort)
- T = 28 temporal tokens per trial (200 Hz $\rightarrow$ 20 Hz patches)
- d = 64 by default, with $d = 128$ as the first width ablation

Cross-patient sharing happens via a single learnable parcel-embedding lookup, not via per-parcel pooling. The lookup table has size $(15, d)$, where the 15 keys are the Tier-1 left-hemisphere Brainnetome parcels (§??). $N_{\text{tok}} = 15$ is therefore the *embedding-lookup table size*, not a token count.

Why this shape, and not the earlier $N_{\text{tok}} \times d \times T$ pooled interface? Three findings drove the rewrite:

1. **Vertex-collision audit (2026-04-16 late).** The strict fsaverage snap collides $\sim 23.6\%$ of electrodes per patient (302/1280 across 7 LH patients), with up to 4 electrodes mapped to the same vertex on the densest 128-channel grids. Per-parcel pooling cannot separate collided electrodes.
2. **μ ECoG vs sEEG compression calculus.** A 15-token atlas pool compresses dense μ ECoG by 22–30 $\times$ per active parcel, vs $\sim 5\times$ for sEEG. Sub-parcel somatotopy (Bouchard 2013, Conant 2018) sits at exactly the scale this pool would discard, and is what the dense device was meant to capture.
3. **Sub-parcel positional alignment is below the noise floor.** Registration noise (~ 3 – 5 mm RMS) plus inter-individual anatomical variability (~ 10 mm RMS) gives ~ 11 mm joint uncertainty, while functional sub-parcel clusters are 5–10 mm wide. SNR for cross-patient sub-parcel positional alignment is ≤ 1 . Any architecture that relies on aligning sub-parcel position across patients is fighting physics.

The B-1 design respects all three: *the rigid μ ECoG grid is the spatial substrate, the parcel is the cross-patient anchor, and sub-parcel positional alignment is not encoded.*

3 Phase-1 Status

The Phase-1 contract is **fixed atlas, supervised only, no learned calibration**. Specifically:

- the Brainnetome PM volume at `data/atlas/BNA_PM.4D.nii.gz` is baked once onto fsaverage (per #36)
- patient electrodes project onto fsaverage via stock `sphere.reg` (snap-to-pial); the `cvs.avg35` cache from the cras-corrected Python port of Zac’s `sub2AvgBrainClinical.m` is retained as a parity oracle only
- no learned gain, offset, rigid correction, parcel offsets, or parcel temperatures

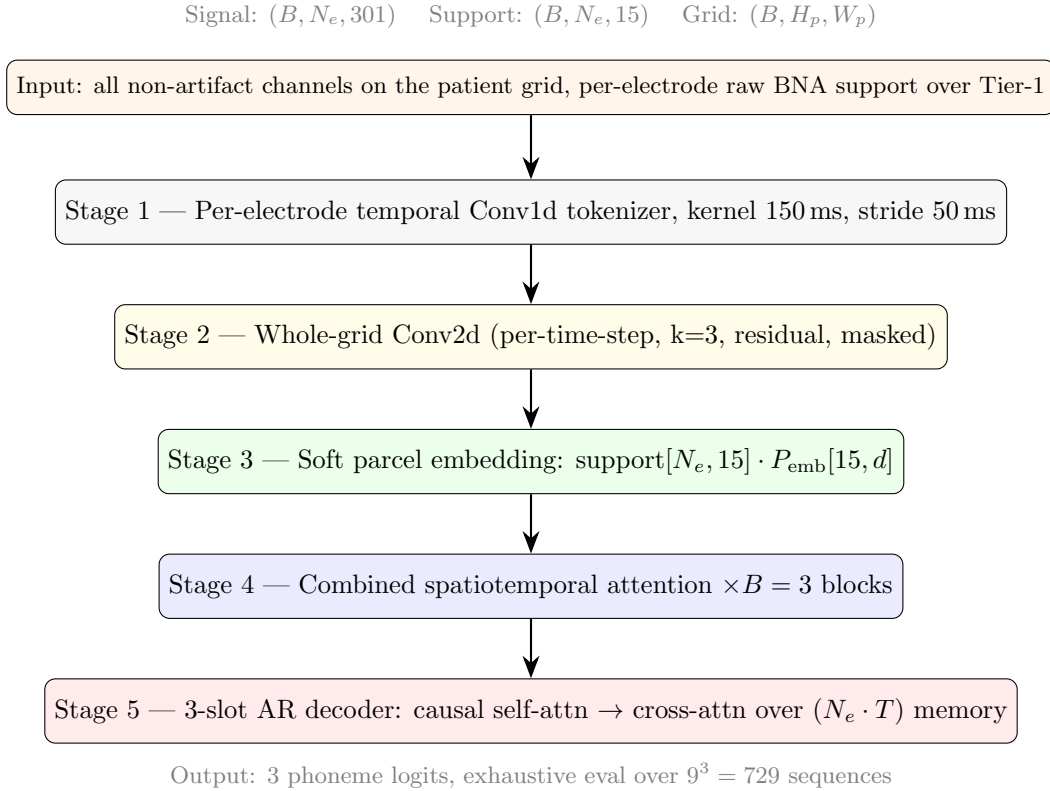
- no MFA in the training path
- no data augmentation in the first benchmark run
- no SSL pretraining, no sEEG, no external datasets

All Phase-1 blockers are closed as of 2026-04-16 late. The two final closures were:

- **#36** — fsaverage spatial pivot adopted after the cras-bug fix in the cvs_avg35 oracle unblocked the parity comparison
- **#34** — phoneme loading audit closed across the full cohort; the `.fif` is the authoritative source for labels, response onsets, and signal, and the BIDS events TSV is a soft cross-check

The v14 loader is unblocked. The remaining work is execution against the frozen contract; the implementation order is in `docs/plans/v14-core.md`.

4 Model Overview



The architecture is intentionally small. Stage 1 extracts short local temporal features per electrode. Stage 2 mixes information across spatially adjacent electrodes on the patient grid. Stage 3 adds the cross-patient parcel-identity prior to each per-electrode token. Stage 4 models region-region and time-time interaction in a single combined attention. Stage 5 reads out a fixed 3-slot phoneme sequence.

5 Spatial Interface

5.1 Coordinates and channel bridge

The stable channel bridge ends at a physical electrode name. Under the fsaverage pivot (#36), this name-keyed bridge terminates in a per-patient surface-projection cache

$$\text{data/fsaverage_coords/}<\text{pt}>_fsaverage_pial.csv, \quad (2)$$

with the cvs_avg35 MNI cache at `data/mni_coords/`<pt>`_MNI152.csv` retained as a parity oracle only. The lookup key is the *physical electrode name* (concatenated from `<pt>.electrodeNames`), not a row index and not a TSV electrode name.

The channel bridge is:

$$\text{.fif channel} \rightarrow \text{amp channel} \rightarrow \text{layout map} \rightarrow \text{physical electrode name} \rightarrow \text{fsaverage vertex} \rightarrow \text{support} \quad (3)$$

The mapping is layout-specific:

- 128-strip patients (S14, S16, S22, S23, S26) use Map 4 from local `*_channelMap.mat`
- 256-grid patients (S33, S39, S62) use Map 3 from `*_channelMapAll.mat`
- S58 uses a cropped Map-3 variant aligned back to the full Map 3
- the stray `S39_channelMap.mat` is non-authoritative and must never be loaded

The bridge contract is frozen (per blocker #12). The patient-by-patient verifier (Phase A2 in `docs/plans/v14-core.md`) also recovers the (row,col) of each electrode on the device grid, which Stage 2 needs to scatter onto the per-patient bounding rectangle.

5.2 Patient-side vs atlas-side surfaces

The two sides of the fsaverage pipeline use different surfaces:

- **Patient side:** project electrodes from each patient's `pial-outer-smoothed` surface through stock `sphere.reg` to fsaverage and snap to the nearest `lh.pial` vertex. This matches the physical fact that the electrodes sit on the dural envelope rather than on the cortical ribbon itself.
- **Atlas side:** bake the Brainnetome PM volume through the true cortical ribbon, from `white` to `pial`, using `mri_vol2surf --projfrac-avg 0 1 0.1` from `ICBM152_fs`, then `mri_surf2surf` to fsaverage, then 2D geodesic surface smoothing at FWHM 3.5 mm. Output cache: `data/atlas/fsaverage_bake_fast2/`.

The PSF is baked into the atlas at bake time. There is no extra query-time Gaussian. Convolving again would double-smooth. The old nearest-neighbor 8 mm dilation bridge belongs to the retired cvs_avg35 path and is not part of the v14 spatial design.

`ICBM152_fs` is not the atlas. It is the FreeSurfer reconstruction of the ICBM152 template, which provides white/pial surfaces in the same space as the BNA PM volume before the nonlinear `sphere.reg`-based transfer to fsaverage.

5.3 Channel inclusion

The input channel policy (per blocker #11) is:

- use **all non-artifact channels**; do not silently filter to significant channels
- remove artifact channels from the active channel axis; do not zero them in place
- significant-channel masks are an ablation-only filter, not the default

5.4 Tier-1 parcels

Phase 1 uses 15 fixed left-hemisphere Brainnetome parcels, derived 2026-04-16 late on the fsaverage bake under the frozen rule

$$\text{argmax_wins}(p) \geq 10, \quad (4)$$

where $\text{argmax_wins}(p)$ is the number of Phase-1 LH electrodes for which p is the dominant parcel across all 246 BNA ROIs. The cohort is 7 LH patients (S14, S16, S23, S26, S33, S39, S62), 1280 electrodes; the ranking source is `data/atlas/fsaverage_parity/fsaverage_cohort_ranking.csv`. The argmax band $[5, 9]$ is empty, so any threshold in $[5, 10]$ yields the same 15 parcels.

The canonical list (and BNA 1-based indices) lives at `src/speech_decoding/v14/token_spec.DEFAULT_BASE_PARCE`

A4hf, A1/2/3ulhf, A1/2/3tonIa, A4tl, IFJ, A44d, A45c, A6cv1, A9/46v,
A40rd, IFS, A2, A40rv, A44v, A22r.

These 15 parcels span face/head and tongue M1/S1, ventral premotor, Broca (dorsal/caudal/ventral), inferior frontal sulcus and junction, supramarginal (rostrorodorsal/rostromedial), area 2 (dorsal S1), and rostral area 22. Under B-1 they serve as the **embedding-lookup keys** for P_{emb} , not as fixed token slots.

5.5 Support and validity

Per-electrode parcel support is read directly from the baked atlas at the electrode’s assigned fsaverage pial vertex (per blocker #5):

$$\text{support}(e, p) = \text{fsaverage_bake}[v_e, p], \quad (5)$$

where v_e is the fsaverage `lh.pial` vertex assigned to electrode e by `src/speech_decoding/v14/fsaverage_project`. Values are raw BNA probabilities (0–100), not row-normalized.

The full $(N_e, 15)$ support matrix is materialized once via `data/atlas/support_cache/<pt>_support_tier1.csv` (Phase A1) and emitted by the loader directly. Electrode validity is **soft** (per blocker #3 amended): support enters as a routing weight, not as a hard membership test. The only hard exclusion is #11 (artifact channels). An electrode whose $\max_{p \in \text{Tier-1}} \text{support}(e, p) = 0$ is still emitted, but its parcel-embedding contribution is zero by construction.

5.6 Loader contract

The v14-core sample dict (per blocker #13 amended) is:

<code>signal</code>	$\in \mathbb{R}^{N_e \times 301}$	float32, productionZscore HGA, 200 Hz, 1.5 s
<code>patient_id</code>	$\in \text{str}$	e.g. "S14"
<code>label</code>	$\in \{0, \dots, 8\}^3$	0-indexed alphabetical ARPABET
<code>electrode_grid_layout</code>	$\in \mathbb{Z}^{N_e \times 2}$	(row, col) on patient grid
<code>electrode_grid_shape</code>	$= (H_p, W_p)$	per-patient bounding rect
<code>electrode_active_mask</code>	$\in \{0, 1\}^{N_e}$	non-artifact and not pad
<code>support</code>	$\in \mathbb{R}^{N_e \times 15}$	raw BNA probability over Tier-1

The fields `coords[N_ch, 3]`, `token_mask[N_tok]`, and `token_support[N_tok]` from the pre-amendment contract are removed. There are no per-parcel tokens, so per-parcel masks are irrelevant; spatial-coordinate PE is dropped because the Conv2d (Stage 2) already encodes local position and that position is not patient-transferable.

6 Stage 1 — Temporal Tokenizer

The temporal front-end is per-electrode, shared across all patients (per blockers #2, #6):

$$\mathbf{x} \in \mathbb{R}^{B \times N_e \times 301} \rightarrow \mathbf{h} \in \mathbb{R}^{B \times N_e \times d \times 28}. \quad (6)$$

For each electrode independently, apply a shared Conv1d patch projection:

- input rate: 200 Hz
- input channels: 1, output channels: $d = 64$
- kernel: 150 ms = 30 samples
- stride: 50 ms = 10 samples
- bias: True
- temporal compression: $10.7\times$, matching the HGA envelope timescale

Output rate is 20 Hz, giving $T = 28$ patches per 1.5 s window. The token semantics are overlapping temporal patches, not continuous framewise states. There is no spatial mixing in Stage 1.

7 Stage 2 — Whole-Grid Conv2d

Stage 2 mixes information across spatially adjacent electrodes on the rigid μ ECoG grid. The output of Stage 1 is scattered onto the per-patient bounding rectangle:

$$\mathbf{h} \in \mathbb{R}^{B \times N_e \times d \times T} \xrightarrow{\text{scatter}} \mathbf{x} \in \mathbb{R}^{B \times d \times H_p \times W_p \times T}, \quad (7)$$

with a per-cell active mask in $\{0, 1\}^{B \times H_p \times W_p}$ (false on pad cells and on artifact channels). Per-patient bounding rectangles are e.g. $H_p \times W_p = 8 \times 16$ for S14, 16×16 for S33, 12×24 for S58.

Per time step, with weights shared across t :

$$\mathbf{x}_{\text{normed}} = \text{LayerNorm}_{\text{channel}}(\mathbf{x}) \quad (8)$$

$$\mathbf{x}_{\text{conv}} = \text{GELU}(\text{Conv2d}(\mathbf{x}_{\text{normed}})) \quad (9)$$

$$\mathbf{x}_{\text{out}} = \mathbf{x} + \mathbf{x}_{\text{conv}} \odot \text{active_mask} \quad (10)$$

with kernel 3×3 , stride 1, padding 1, full channel mixing, pre-norm LayerNorm, residual masked by the active mask. The default depth is one layer; depth 2 is ablation A3.

Justifications:

- **Receptive field.** A 3×3 kernel covers $\pm 1.33\text{--}3\text{ mm}$ at typical μECoG pitch, which matches the HGA spatial correlation length.
- **Full channel mixing.** $64 \times 64 \times 9 \approx 37\text{K}$ parameters per layer is trivial at our scale; depthwise-separable saves nothing.
- **Per-time-step.** Temporal mixing belongs to Stage 4. Stage 2 is a pure spatial mixer.
- **Mask handling.** Pad cells and artifact channels are zeroed from the residual contribution, so they do not bleed through the conv.

After Stage 2, gather back to the per-electrode layout: $(B, d, H_p, W_p, T) \rightarrow (B, N_e, d, T)$.

8 Stage 3 — Soft Parcel Embedding

Stage 3 adds the only cross-patient anatomical signal in the model. For each electrode e , the parcel-identity contribution is a support-weighted sum over a learnable lookup table:

$$\text{parcel_emb}(e) = \sum_{p \in \text{Tier-1}} \text{support}(e, p) \cdot P_{\text{emb}}[p], \quad P_{\text{emb}} \in \mathbb{R}^{15 \times d}. \quad (11)$$

P_{emb} is Xavier-uniform initialized and has zero per-patient parameters. Support is consumed raw (not row-normalized), so electrodes outside Tier-1 contribute weak embeddings naturally and electrodes split across two parcels get a convex-like combination of both embeddings.

The Stage-3 token, broadcast across time, is:

$$\text{token}(e, t) = \mathbf{h}_{\text{conv}}(e, t) + \text{parcel_emb}(e). \quad (12)$$

Flatten to the backbone input:

$$\mathbf{z} \in \mathbb{R}^{B \times (N_e \cdot T) \times d}, \quad \text{token_active_mask} \in \{0, 1\}^{B \times (N_e \cdot T)}, \quad (13)$$

where the active mask is `electrode_active_mask` broadcast across T .

What this token deliberately contains:

- sub-parcel functional information: implicit in $\mathbf{h}_{\text{conv}}$ via Stage 2 grid mixing
- parcel identity: explicit via the soft `parcel_emb`, the cross-patient anchor
- boundary-awareness: a convex combination for electrodes that straddle two Tier-1 parcels

What it deliberately does not contain:

- grid-coordinate embedding (Stage 2 already encodes local position; not transferable across patients)
- intra-parcel coordinate (PE-2 / PE-3 / parcel frames) — registration noise dominates anatomical variability, so cross-patient sub-parcel alignment is below the SNR floor
- fsaverage vertex coordinate — collision-prone and not transferable
- hard parcel argmax — replaced by soft probability weighting

The within-parcel Perceiver summarizer (the pre-amendment design) is retired. The parcel-frame chart and `parcel_frames.npz` are retired. The mean+gradient pool is the main linear ablation (A5), not a pipeline element.

9 Stage 4 — Combined Spatiotemporal Backbone

After Stage 3 the model operates on a single per-electrode token stream:

$$\mathbf{z} \in \mathbb{R}^{B \times (N_e \cdot T) \times d}, \quad N_e \in \{128, 256\}, \quad T = 28. \quad (14)$$

The backbone is $B = 3$ **combined** attention blocks. Each block is one combined-attention layer over all $(N_e \cdot T)$ tokens, followed by an FFN:

$$\mathbf{q} = \text{LayerNorm}(\mathbf{z}) \quad (15)$$

$$\mathbf{z} \leftarrow \mathbf{z} + \text{MultiHeadAttention}(\mathbf{q}, \mathbf{q}, \mathbf{q}; \text{mask}, \text{RoPE}_{\text{temp}}) \quad (16)$$

$$\mathbf{z} \leftarrow \mathbf{z} + \text{FFN}(\text{LayerNorm}(\mathbf{z})) \quad (17)$$

Hyperparameters:

- num_heads = 4, head_dim = 16 (so num_heads · head_dim = $d = 64$)
- FFN: $\text{Linear}(d \rightarrow 4d) \rightarrow \text{GELU} \rightarrow \text{Linear}(4d \rightarrow d)$, hidden width $4d = 256$
- pre-norm, dropout 0.1
- RoPE on the temporal axis only: rotation depends on $t_a - t_b$ and is identity for token pairs at the same t
- attention mask: `token_active_mask` broadcast to pairs, applied on both key and query axes
- inactive query rows zeroed after each block (storage hygiene; the mask carries the semantics)

Why combined and not factored:

- **Compute is not binding.** At the worst-case 256-channel patient, $N_e \cdot T = 7168$ tokens, combined attention is ~ 13 B FLOPs per layer and ~ 80 B per forward, well within budget.
- **Combined captures lag-coupled cross-region patterns in one layer.** Speech production has planning $\rightarrow$ execution $\rightarrow$ feedback patterns that are intrinsically cross-spatial-temporal. Factored attention requires a 2-layer composition to express the same $(e_i, t_a) \rightarrow (e_j, t_b)$ interaction.

- **$B = 3$ blocks (down from 6 factored half-blocks).** Each combined block is more expressive, fewer are needed.
- **No spatial PE.** Spatial relationships emerge from data plus the soft parcel embedding plus the Stage 2 conv. The temporal axis is the only one with an explicit positional prior.

No SC/FC bias in the baseline. The structural- and functional-connectivity additive logit bias (originally in #8) is deferred to ablation A4. When reintroduced it will be per-(layer, head) learnable scalars α_{SC}, α_{FC} over a precomputed $\text{support} \cdot \text{SC}_{\text{norm}} \cdot \text{support}^\top$ parcel-pair bias, broadcast across temporal pairs.

The full $(N_e \cdot T, d)$ backbone output is handed to the decoder. There is no early pooling.

10 Stage 5 — Decoder and Supervision

The target is a fixed 3-slot phoneme sequence over the 9 canonical ARPABET classes, alphabetically indexed (per blocker #16):

$$\text{AA} = 0, \text{AE} = 1, \text{B} = 2, \text{G} = 3, \text{IY} = 4, \text{K} = 5, \text{P} = 6, \text{UW} = 7, \text{V} = 8. \quad (18)$$

The decoder (per blocker #28) is one small autoregressive block:

- one learned base query, plus one slot embedding per position (3 slots)
- previous-token embedding injected into the query stream
- one causal self-attention layer across the 3 slots
- one cross-attention layer over the flattened backbone memory $M \in \mathbb{R}^{(N_e \cdot T) \times d}$
- one shared linear vocab head

Slot k computes

$$q'_k = q_k + \text{emb}(y_{k-1}), \quad u_k = \text{CausalSelfAttn}(q'_k), \quad o_k = \text{CrossAttn}(u_k, M), \quad \hat{y}_k = W o_k. \quad (19)$$

Cross-attention naturally handles the variable $N_e \cdot T$ memory size across patients.

10.1 Loss and decode

The baseline supervised loss is plain slot-wise cross-entropy (per blocker #9):

$$\mathcal{L} = \frac{1}{3} \sum_{k=1}^3 \text{CE}(\hat{y}_k, y_k). \quad (20)$$

There is no label smoothing, focal loss, mixup, or auxiliary token head in the first benchmark run. Training uses teacher forcing.

Evaluation uses exhaustive decoding over all $9^3 = 729$ possible sequences. Greedy decoding is allowed as a comparison path only. The reported metric is slot-averaged phoneme error rate (PER).

11 Hyperparameter Summary

Stage	Field	Value
Stage 1 (Conv1d)	in/out channels	1 / 64
	kernel / stride	30 samples (150 ms) / 10 samples (50 ms)
	output rate	20 Hz, $T = 28$
Stage 2 (Conv2d)	in/out channels	64 / 64
	kernel / stride / pad	3 / 1 / 1
	channel mixing	full
	num.layers	1 (depth 2 = ablation A3)
	norm / activation	pre-norm LayerNorm / GELU
	residual	masked by <code>electrode_active_mask</code>
Stage 3 (Embed)	P_{emb} shape	(15, 64)
	init	Xavier uniform
	support	raw (not normalized)
	Tier-1 source	<code>token_spec.DEFAULT_BASE_PARCELS</code>
Stage 4 (Backbone)	num.blocks	3
	d_{model}	64 (128 = ablation A1)
	num.heads / head.dim	4 / 16
	FFN width / activation	256 = $4d$ / GELU
	norm / dropout	pre-norm LayerNorm / 0.1
	attention	combined spatiotemporal
	RoPE axis	temporal only
	SC/FC bias	none in baseline (ablation A4)
Stage 5 (Decoder)	num.slots	3
	self-attn / cross-attn layers	1 / 1
	vocab head	shared linear, 9 phonemes
	training / eval	teacher-forced / exhaustive 9^3

12 Training and Evaluation Defaults

These are not architecture innovations. They are frozen first-run defaults chosen to prevent protocol drift.

12.1 Evaluation

- grouped-by-token cross-validation (per blocker #33)
- 5 outer folds
- 3 seeds: 42, 137, 256
- 20% grouped validation split within each training fold
- same train/val partition reused across seeds within fold
- report per-patient PER mean $\pm$ std across seeds, then population mean

12.2 Training

- optimizer: AdamW

- base learning rate: 10^{-3}
- weight decay: 10^{-4}
- scheduler: cosine decay
- warmup: 20 epochs
- gradient clipping: 1.0
- mixed precision on by default on DCC unless unstable
- early stopping on validation PER

12.3 Batching

- grouped-by-patient batches (per blocker #31)
- baseline run is per-patient first
- `trials_per_batch = 8`
- effective batch target 32 via gradient accumulation if needed
- max epochs 300
- patience 10 validation checks
- validation every 5 epochs

13 Ablation Plan

Once `v14-core` is running and overfits a tiny subset cleanly, the following ablations execute in priority order (Phase F):

1. **A3 (depth-2 GridMixer)** — Stage 2 `num_layers = 2`
2. **A1** ($d_{\text{model}} = 128$) — width sweep
3. **A2 (block count)** — $B \in \{2, 4\}$ vs $B = 3$ baseline
4. **A4 (SC/FC bias)** — per-(layer, head) $\alpha_{\text{SC}}, \alpha_{\text{FC}}$ over a parcel-pair bias
5. **A5 (mean+gradient linear pool)** — main linear comparator to the soft embedding
6. **A6 (factored attention head-to-head)** — factored as historical reference
7. **A7 (intra-parcel PE-2)** — patient-grid bbox PE; tests whether explicit position helps despite the alignment-noise argument

14 What Is Deferred

The following are explicitly not part of the Phase-1 base architecture:

- learned per-patient calibration (Δ/ω , δ_l , τ_l , additional gain/offset)

- SSL pretraining
- sEEG
- right-hemisphere integration (S22, S58)
- external datasets (Flinker, Chang chronic ECoG)
- broad recipe ablations (label smoothing, focal loss, mixup, augmentation)
- articulatory or auxiliary heads

For future external datasets with uncertain MNI quality, a reasonable Phase-2 fallback is:

1. freeze the shared neural network weights
2. fit a per-patient rigid-body coordinate adapter only
3. restrict that adapter to rotation plus translation, not free affine warping

That should be treated as an explicit external-dataset adaptation step, not as part of the base Phase-1 contract.

15 Sequencing

- **Phase 1:** supervised `v14-core` on response-locked μ ECoG, 4 core LH patients, 5 folds $\times$ 3 seeds = 60 jobs on DCC
- **Phase 1.5:** SSL on the full continuous μ ECoG corpus (not response-locked-only)
- **Phase 2:** learned per-patient calibration on top of the fixed-atlas baseline
- **Phase 3+:** sEEG join, external datasets, broader scaling

The current state is: all blockers closed, the contract is frozen, and the work is implementation against `docs/plans/v14-core.md`. Performance against the historical 0.734 ± 0.007 S14 baseline is the next conversation, after correctness lands.

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